Removing the Sigma-Finite Hypothesis in an L_1-Amalgamation Lemma

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2026-06-21

Source and Status

This packet answers a specific question in Antonio Aviles and Pedro Tradacete, *Amalgamation and injectivity in Banach lattices*, arXiv:2007.15261. The status is

`full_solution_likely_valid`

for the sigma-finiteness-removal question after Lemma 4.2.

Source Question

Lemma 4.2 of the source paper proves the following L_1-amalgamation result under the assumption that the common measure ν_0 is σ-finite: if

$$u_i : L_1(\nu_0) \longrightarrow L_1(\nu_i), \qquad i = 1, 2,$$

are lattice isometric embeddings, then there are a positive measure ν_3 and lattice isometric embeddings

$$v_i : L_1(\nu_i) \longrightarrow L_1(\nu_3), \qquad i = 1, 2,$$

such that $v_1u_1 = v_2u_2$.

but by (1) above, and the definition of ν_{0i}, this implies that

$$\begin{aligned}\nu_3(g_i^{-1}(A)) &= \nu_{012}\{(x_0,x_1,x_2) : x_0 = f_i(x_i), x_i \in A\}\\ &= \nu_{012}^{\{0,i\}}\{(x_0,x_i) : x_0 = f_i(x_i), x_i \in A\}\\ &= \nu_{0i}\{(x_0,x_i) : x_0 = f_i(x_i), x_i \in A\}\\ &= \nu_i(A).\end{aligned}$$

□

As a side remark, Theorem 4.1 is also true if we change everywhere *positive measure* by *signed measure*, just by calling Theorem 2.2 instead of Corollary 2.4.

Lemma 4.2. *Let ν_0,ν_1,ν_2 be positive measures such that ν_0 is σ-finite. If we have isometric lattice embeddings $u_1 : L_1(\nu_0) \longrightarrow L_1(\nu_1)$ and $u_2 : L_1(\nu_0) \longrightarrow L_1(\nu_2)$, then we can find another positive measure ν_3 and isometric lattice embeddings $v_1 : L_1(\nu_1) \longrightarrow L_1(\nu_3)$ and $v_2 : L_1(\nu_2) \longrightarrow L_1(\nu_3)$ with $v_1 \circ u_1 = v_2 \circ u_2$.*

Proof. As a first case, we suppose that all measures ν_0,ν_1,ν_2 are finite and the embeddings u_1,u_2 are measure preserving. In that case, a result of Iwanik [14] provides, for $i = 1,2$ measure-preserving $f_i : \Omega_i \longrightarrow \Omega_0$ such that $u_i(h) = h \circ f_i$ for all $h \in L_1(\nu_i)$. Although Iwanik's results require the measure space to be homeomorphic to a Borel subset of the Hilbert cube, Maharam's theorem allows us to apply them here without loss of generality. We can then invoke Theorem 4.1 and define $v_i(h) = g_i \circ h$.

We consider now the general case. By Lemma 3.1, we can suppose that ν_0 is a probability measure. Let $\mathbb{I}_0 \in L_1(\nu_0)$ be the constant function equal to 1. For $i = 1,2$, we consider $\mathbb{I}_i = u_i(\mathbb{I}_0)$, S_i the support of $\mathbb{I}_i$, the measure $\tilde{\nu}_i$ on S_i given by $\tilde{\nu}_i(B) = \int_B \mathbb{I}_i \, d\nu_i$, and $\tilde{u}_i : L_1(\nu_0) \longrightarrow L_1(\tilde{\nu}_i)$ given by

$$\tilde{u}_i(h) = \left.\frac{u_i(h)}{\mathbb{I}_i}\right|_{S_i}.$$

The measures $\nu_0,\tilde{\nu}_1,\tilde{\nu}_2$ are finite (in fact probability measures), and $\tilde{u}_1,\tilde{u}_2$ are measure preserving isometric embeddings. So we can apply the first case considered in this proof, and we obtain lattice isometric embeddings $\tilde{v}_i : L_1(\tilde{\nu}_i) \longrightarrow L_1(\tilde{\nu}_3)$ with $\tilde{v}_1 \circ \tilde{u}_1 = \tilde{v}_2 \circ \tilde{u}_2$. Finally, we take

$$L_1(\nu_3) = L_1(\tilde{\nu}_3) \oplus_1 L_1(\nu_1|_{\Omega_1\setminus S_1}) \oplus_1 L_1(\nu_2|_{\Omega_1\setminus S_2})$$

and $v_i : L_1(\nu_i) \longrightarrow L_1(\nu_3)$ as

$$\begin{aligned}v_1(h) &= \tilde{v}_1\left(\left.\frac{h}{\mathbb{I}_1}\right|_{S_1}\right) \oplus h|_{\Omega_1\setminus S_1} \oplus 0,\\ v_2(h) &= \tilde{v}_2\left(\left.\frac{h}{\mathbb{I}_2}\right|_{S_2}\right) \oplus 0 \oplus h|_{\Omega_2\setminus S_2}.\end{aligned}$$

Immediately after the lemma, the authors ask whether the technical σ-finiteness assumption on ν_0 can be removed.

□

We do not know if the technical condition that ν_0 is σ-finite can be removed from Lemma 4.2.

In the remaining of the section, we will focus on the push-out construction for Banach lattices. Before introducing this, we need to recall the following tool: Given a Banach space E, the *free Banach lattice generated by* E is a Banach lattice $FBL[E]$ together with a linear isometric embedding $\delta : E \to FBL[E]$ with the property that for every Banach lattice X and every operator $T : E \to X$, there is a unique lattice homomorphism $\hat{T} : FBL[E] \to X$ such that $T = \hat{T} \circ \delta$, and moreover $\|\hat{T}\| = \|T\|$. This notion was introduced in [4], extending an earlier construction of free Banach lattices over a set of generators given in [8].

In categorical terms, for objects A_0, A_1, A_2 and morphisms $\alpha_i : A_0 \to A_i$, $i = 1, 2$, a push-out diagram is an object $PO = PO(\alpha_1, \alpha_2)$ together with morphisms $\beta_i : A_i \to PO$, $i = 1, 2$, making commutative the diagram

$$\begin{tikzcd}
A_1 \arrow[r, "\beta_1"] & PO \\
A_0 \arrow[u, "\alpha_1"] \arrow[r, "\alpha_2"] & A_2 \arrow[u, "\beta_2"]
\end{tikzcd}$$

and with the universal property that if $\beta_i' : A_i \to B$ are such that $\beta_1'\alpha_1 = \beta_2'\alpha_2$, then there is a unique $\gamma : PO \to B$ such that $\gamma\beta_i = \beta_i'$ for $i = 1, 2$, as the following diagram illustrates:

$$\begin{tikzcd}
 & & B \\
A_1 \arrow[r, "\beta_1"] \arrow[urr, bend left, "\beta_1'"] & PO \arrow[ur, "\gamma"] & \\
A_0 \arrow[u, "\alpha_1"] \arrow[r, "\alpha_2"] & A_2 \arrow[u, "\beta_2"] \arrow[uur, bend right, "\beta_2'"'] &
\end{tikzcd}$$

In the case of Banach lattices, the above definition may be called an isomorphic push-out. The definition of the isometric push-out adds the extra condition that $\max\{\|\beta_1\|, \|\beta_2\|\} \leq 1$, and in the universal property we have that $\|\gamma\| \leq \max\{\|\beta_1'\|, \|\beta_2'\|\}$. It is an easy exercise, using the universal property, that the isomorphic push-out is uniquely determined up to lattice isomorphism, while the isometric push-out is uniquely determined up to lattice isometry. In the sequel, we will always consider isometric push-outs.

Let us see how to make the push-out construction in the category $\mathcal{BL}$: Given Banach lattices X_0, X_1, X_2 and lattice homomorphisms $T_i : X_0 \to X_i$ for $i = 1, 2$, let $X_1 \oplus_1 X_2$ denote the direct sum equipped with the norm $\|(x_1, x_2)\| = \|x_1\| + \|x_2\|$ and let $j_i : X_i \to X_1 \oplus_1 X_2$ denote the canonical embedding for $i = 1, 2$. Let Z be the (closed) ideal in $FBL[X_1 \oplus_1 X_2]$ generated by the families $(\delta(j_1|x|) -$

Idea of the Proof

The only obstruction to directly applying Lemma 4.2 is that the common $L_1(\nu_0)$ may have no weak unit. Maharam's theorem decomposes any L_1 space as an ℓ_1-sum of standard pieces, each of which is σ-finite: one-dimensional atomic pieces and finite-measure product pieces $L_1([0,1]^\kappa)$. We apply the source lemma on each component.

Because lattice homomorphisms preserve disjointness, the images of distinct components generate pairwise disjoint bands in each codomain. After amalgamating component by component, we take the ℓ_1-sum of the component amalgams and add the two residual codomain bands. This sum is again an L_1 space, and it gives the desired global amalgamation.

measure of the Lebesgue measure on each factor. If τ is a cardinal and X a Banach lattice, $\ell_1(\tau, X)$ will be the ℓ_1-sum of τ many copies of X.

Theorem 3.4 (Maharam)**.** *Every Banach lattice of the form $L_1(\mu)$ is lattice isometric to an ℓ_1-sum of Banach lattices of the form $\ell_1(\Gamma)$ or $L_1([0,1]^\kappa)$,*

$$L_1(\mu) \simeq \ell_1(\Gamma) \oplus_1 \left(\bigoplus_{i\in I} \ell_1(\tau_i, L_1([0,1]^{\kappa_i})) \right)_{\ell_1}.$$

The above expression is actually unique if the cardinals κ_i and τ_i are always taken to be either 1 or uncountable. This restriction is necessary because if $1 \leq \alpha \leq \aleph_0$ then $L_1([0,1]) \simeq L_1([0,1]^\alpha)$ and $L_1([0,1]^\kappa) \simeq \ell_1(\alpha, L_1([0,1]^\kappa))$, cf. [20, Theorem 14.10]. Sometimes the products $\{0,1\}^\kappa$ of the $\frac{1}{2}$-discrete measure on $\{0,1\}$ are considered instead of $[0,1]^\kappa$, but this is irrelevant since $L_1([0,1]^\kappa) \simeq L_1(\{0,1\}^\kappa)$ when κ is infinite.

For the following elementary lemma it is more convenient to write Maharam's decomposition without grouping summands by cardinalities. All cardinals κ_i and κ'_j are assumed to be nonzero.

Lemma 3.5. *Suppose that we have two L_1 spaces that can be written in the form*

$$L_1(\mu) \simeq \left(\bigoplus_{i\in I} L_1([0,1]^{\kappa_i}) \right)_{\ell_1}, \quad L_1(\nu) \simeq \left(\bigoplus_{j\in J} L_1([0,1]^{\kappa'_j}) \right)_{\ell_1}$$

in such a way that $|J| \leq |I|$ and $\sup_{j\in J} \kappa'_j \leq \min_{i\in I} \kappa_i$. Then, for every separable sublattice $X \subset L_1(\mu)$, there exists $Y \simeq L_1(\nu)$ such that $X \subset Y \subset L_1(\mu)$.

Proof. We can suppose that J is infinite because, as we mentioned before, we have that $L_1([0,1]^\kappa) \simeq \ell_1(\aleph_0, L_1([0,1]^\kappa)$. Every element $f \in L_1(\mu)$ is supported on countably many summands of the ℓ_1-sum. Moreover, each nonzero component $f_i \in L_1([0,1]^{\kappa_i})$ of f depends on countably many coordinates of the cube. Since X is separable, this means that we have

$$X \subset \left(\bigoplus_{i\in I_0} L_1([0,1]^{A_i}) \right)_{\ell_1} \subset L_1(\mu),$$

where I_0 and $A_i \subset \kappa_i$ are countable sets. Here we identify $L_1([0,1]^{A_i})$ as the set of all $g \in L_1([0,1]^{\kappa_i})$ that (some representative) depend only on coordinates from A_i, in the sense that $x|_{A_i} = y|_{A_i}$ implies $g(x) = g(y)$. By enlarging the set I_0 and taking large enough sets A'_i for $i \in I_0$ we get a copy of $L_1(\nu)$ as required. $\square$

4. Amalgamation of measure spaces and Banach lattices

A function $f : \Omega \longrightarrow \Omega'$ between measure space (Ω, ν) and (Ω', ν') is measure-preserving if it is measurable and $\nu'(A) = \nu(f^{-1}(A))$ for all $A \in \Sigma'$. As a consequence of Kellerer's marginal problem, we can stablish the following property of measure-preserving maps, which we will refer to as backwards-amalgamation:

Main Result

Theorem. *Let ν_0, ν_1, ν_2 be arbitrary positive measures. If*

$$u_i : L_1(\nu_0) \longrightarrow L_1(\nu_i), \qquad i = 1, 2,$$

are lattice isometric embeddings, then there exist a positive measure ν_3 and lattice isometric embeddings

$$v_i : L_1(\nu_i) \longrightarrow L_1(\nu_3), \qquad i = 1, 2,$$

such that

$$v_1 u_1 = v_2 u_2.$$

Thus the σ-finiteness hypothesis on ν_0 in Lemma 4.2 of the source paper is unnecessary.

Proof. We use standard facts about abstract L_1 spaces: bands generated by pairwise disjoint positive elements decompose as ℓ_1-sums, principal bands generated by weak units are L_1 spaces over σ-finite measures, and ℓ_1-sums of L_1 spaces are again L_1 spaces over the disjoint union measure.

By Maharam's theorem, in the form quoted as Theorem 3.4 in the source paper, we may write

$$L_1(\nu_0) = \left(\bigoplus_{\alpha \in A} E_\alpha \right)_{\ell_1},$$

where each E_α is lattice isometric either to a one-dimensional atomic L_1 space or to $L_1([0,1]^{\kappa_\alpha})$ with its product probability measure. In particular, every E_α is an L_1 space over a σ-finite measure and has a positive weak unit e_α.

Fix $i = 1, 2$. Since u_i is a lattice homomorphism, the positive elements

$$b_{i,\alpha} = u_i(e_\alpha), \qquad \alpha \in A,$$

are pairwise disjoint in $L_1(\nu_i)$. Let $F_{i,\alpha}$ be the principal band generated by $b_{i,\alpha}$. Then the bands $(F_{i,\alpha})_{\alpha \in A}$ are pairwise disjoint. Also

$$u_i(E_\alpha) \subset F_{i,\alpha}.$$

Indeed, if $x \in E_\alpha$, then $|x|$ belongs to the principal band generated by e_α; equivalently,

$$|x| = \lim_{n \to \infty} (|x| \wedge n e_\alpha)$$

in the L_1-norm. Applying the lattice homomorphism u_i gives

$$|u_i x| = u_i |x| = \lim_{n \to \infty} u_i(|x| \wedge n e_\alpha),$$

and each term is dominated by $n b_{i,\alpha}$. Hence $u_i x$ lies in the principal band $F_{i,\alpha}$.

Let $F_{i,0}$ be the band complement of the band generated by all $(F_{i,\alpha})_{\alpha \in A}$. Then

$$L_1(\nu_i) = \left(\bigoplus_{\alpha \in A} F_{i,\alpha} \right)_{\ell_1} \oplus_1 F_{i,0}.$$

Each $F_{i,\alpha}$ is itself an L_1 space, and the restricted maps

$$u_{i,\alpha} : E_\alpha \longrightarrow F_{i,\alpha}$$

are lattice isometric embeddings.

Now apply Lemma 4.2 of the source paper to the pair

$$u_{1,\alpha} : E_\alpha \to F_{1,\alpha}, \qquad u_{2,\alpha} : E_\alpha \to F_{2,\alpha}.$$

The common domain E_α is σ-finite, so the lemma gives an L_1 space W_α and lattice isometric embeddings

$$r_{i,\alpha} : F_{i,\alpha} \longrightarrow W_\alpha, \qquad i = 1, 2,$$

such that

$$r_{1,\alpha} u_{1,\alpha} = r_{2,\alpha} u_{2,\alpha}.$$

Define

$$W = \left(\bigoplus_{\alpha \in A} W_\alpha \right)_{\ell_1} \oplus_1 F_{1,0} \oplus_1 F_{2,0}.$$

This is an abstract L_1 space, hence lattice isometric to $L_1(\nu_3)$ for some positive measure ν_3. Define $v_1 : L_1(\nu_1) \to W$ by

$$v_1 \left((y_\alpha)_{\alpha \in A} \oplus y_0 \right) = (r_{1,\alpha} y_\alpha)_{\alpha \in A} \oplus y_0 \oplus 0,$$

and define $v_2 : L_1(\nu_2) \to W$ by

$$v_2 \left((z_\alpha)_{\alpha \in A} \oplus z_0 \right) = (r_{2,\alpha} z_\alpha)_{\alpha \in A} \oplus 0 \oplus z_0.$$

Both maps are lattice isometric embeddings.

Finally, if $x = (x_\alpha)_{\alpha \in A} \in L_1(\nu_0)$, then

$$u_i x = (u_{i,\alpha} x_\alpha)_{\alpha \in A} \oplus 0.$$

Therefore

$$v_1 u_1 x = (r_{1,\alpha} u_{1,\alpha} x_\alpha)_{\alpha \in A} \oplus 0 \oplus 0 = (r_{2,\alpha} u_{2,\alpha} x_\alpha)_{\alpha \in A} \oplus 0 \oplus 0 = v_2 u_2 x.$$

This completes the amalgamation without assuming that ν_0 is σ-finite. $\square$

Scope and Limitations

This solves only the local question after Lemma 4.2: the L_1-amalgamation lemma remains true for arbitrary common $L_1(\nu_0)$. It does not resolve the later question in the paper about whether every separably $\mathcal{BL}$-injective Banach lattice is isomorphic to a 1-separably $\mathcal{BL}$-injective one.

Novelty Check

Before promotion, the run indexes were searched for arXiv:2007.15261, the title, "Banach lattice amalgamation", "injective Banach lattice", and related amalgamation keywords. The only hits were broad amalgamation packets for other papers and categories, not this L_1-amalgamation condition. External web search for the exact phrases around the sigma-finiteness question and for later references to the paper found the source paper but no later answer to this specific question.

References

- Antonio Aviles and Pedro Tradacete, *Amalgamation and injectivity in Banach lattices*, arXiv:2007.15261.
- H. E. Lacey, *The isometric theory of classical Banach spaces*, used through the Maharam classification quoted in the source paper.

Human Review Recommendation

Review the standard band-decomposition step in the codomain L_1 spaces: the proof uses that pairwise disjoint principal bands in an abstract L_1 space form an ℓ_1-summand together with their band complement. With that standard fact, the componentwise use of Lemma 4.2 gives a complete answer to the source question.